Analyzing Smart City Bike Sharing Data Using Power BI

1.Project Overview:

This project explores a comprehensive dataset of bike-sharing systems spanning **25 different cities** with data from **2,855 unique stations**. The dataset provides a "snapshot" of the bike-sharing infrastructure, capturing key metrics that define the service's reliability and reach.

2.Project Objective:

The primary objective of this project is to analyze the operational efficiency and accessibility of urban bike-sharing networks across multiple international cities.

To analyse which city has high and low revenue

To find out how the revenue changed over the years.

Identify the underperforming cities with low cost efficiency.

3.Attributes:

Attribute Name	Description	Data Type
Number	A unique identification number assigned to each bike station.	*Integer*
Station Name	The official name or label of the station, often including the station number and location.	String
Address	The physical street address or descriptive location where the station is situated.	String
Position	Latitude and longitude coordinates representing the exact geographic location of the station.	String/GPS
Banking	Indicates if the station is equipped with a payment terminal (True = Has terminal, False = No terminal).	Boolean
Bonus	Indicates if the station is a "bonus" station (often used in incentive programs	Boolean

	where users get extra time for returning bikes here).	
Status	The current operational state of the station (e.g., "OPEN" or "CLOSED").	String
Contract Name	The name of the city where the bike-sharing station is located.	String
Bike stands	The total number of docking points (capacity) available at the station.	Integer
Available bike stands	The number of empty docking points currently available to return a bike.	Integer
Available Bike	The number of functional bikes currently docked and available for rental.	Integer
Last Update	The most recent timestamp when the station's status or availability was refreshed in the system.	Date time

4.Data Pre-Processing:

Using Excel:

Creating Fact and dimensional table using **vlookup function**.The three dimensional tables are

1.STATION_DIM

2.DATE_DIM

3.CITY_DIM

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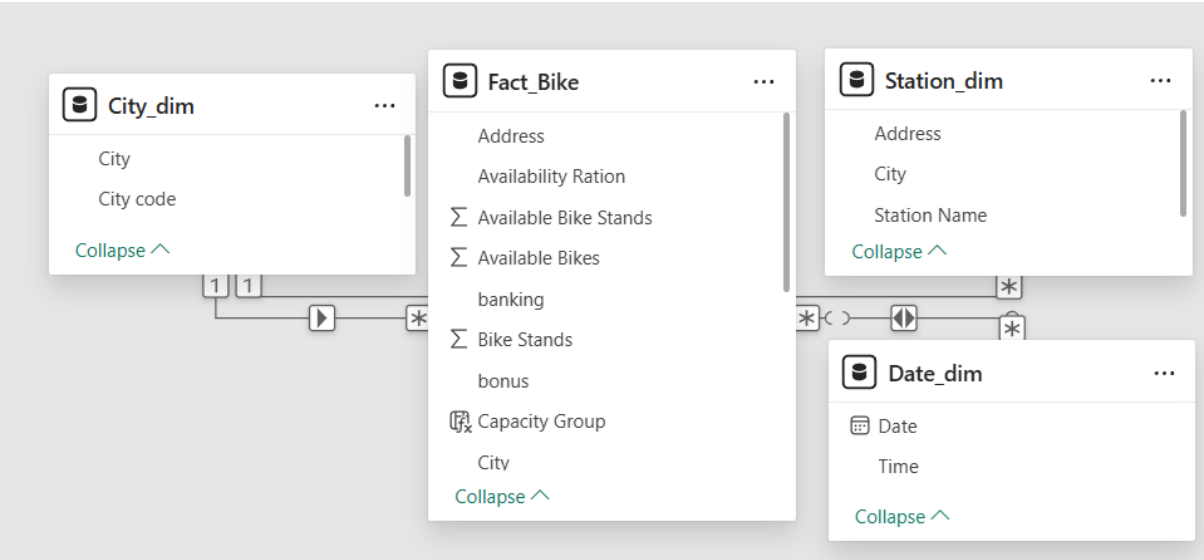

Using Power BI:

Data cleaning and transformation:

Changing the column names into proper format,where number as Number,station name as Station Name,Last updated as Date Time and contract name as City.

Apply clean,trim and capitalize each word functions to City column.Apply Clean and Trim the Station name and Address columns and use **fill up** to the missed address values.

Split the Date Time column into two columns Date and Time.

Change the appropriate Data type to every column.

Creating calculated columns:

Availability Ratio=Divide([Available Bike],[Bike Stand])

Creating conditional column as,

Empty Station=IF([Available Bikes] = 0, "Yes", "No")

Full Station=if[Available bike stands]=0,”Yes”,”No”)

Capacity Group =Switch(True(),[BikeStand]<=20,”Small”,[Bike Stand]<=40,”/Medium”,”Large”)

Station Revenue = (Fact_Bike[Bike Stands]-Fact_Bike[Available Bikes])*3

City Revenue = calculate(sum(Fact_Bike[Station Revenue]),ALLEXCEPT(Fact_Bike,Fact_Bike[City]))

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Creating Measures:

Total Bike Available = sum(Fact_Bike[Available Bikes])

Total Capacity = sum(Fact_Bike[Bike Stands])

Bike Utilization = DIVIDE([Total Bike Available],[Total Capacity])

Total Station = DISTINCTCOUNT(Fact_Bike[Station Name])

Average_Bike_perStation = average(Fact_Bike[Available Bikes])

Estimated revenue = ([total bike stand]-[Total Bike Available])*3

5.Analysis and Visualization:

Using dashboard features,multiple visualization based on problem statement.

City Revenue To analyse which cities earned the highest and lowest revenue.

City separated by capacity group:

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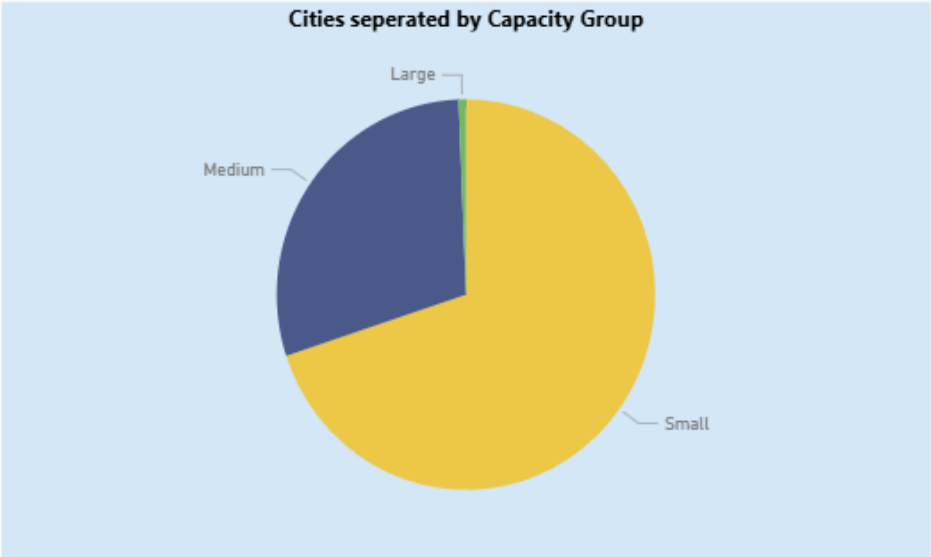
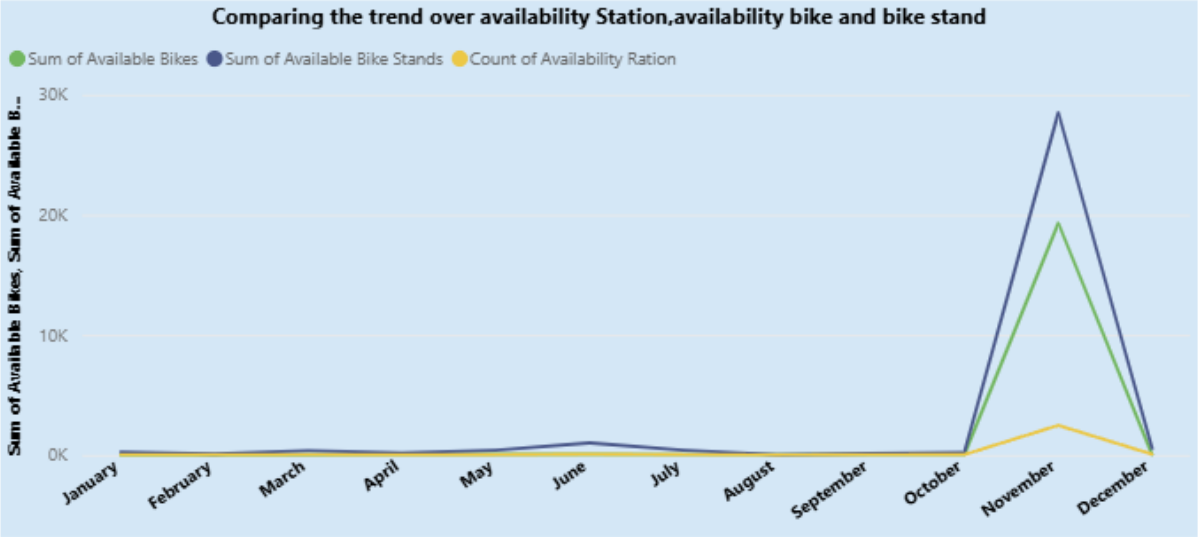
To analyse how the capacity group separated using pie chart.



Comparing the trend over available bike,bike stand and station:

Identify and compare the available bike,bike stand,station over the timeline(month)



Underperforming city with low efficiency:

Using scatter plot identify the utilization range over city

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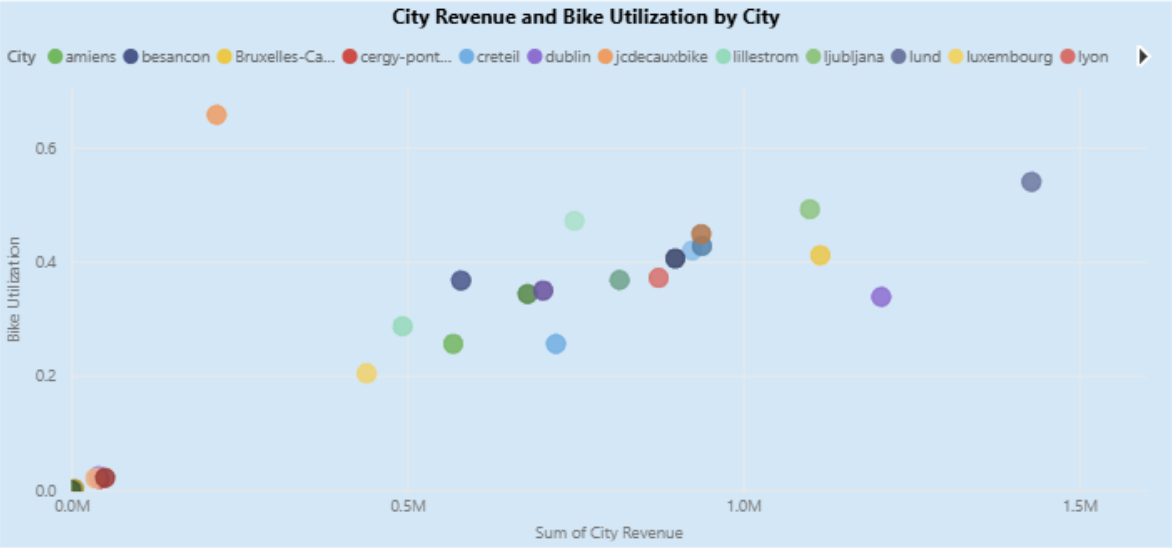
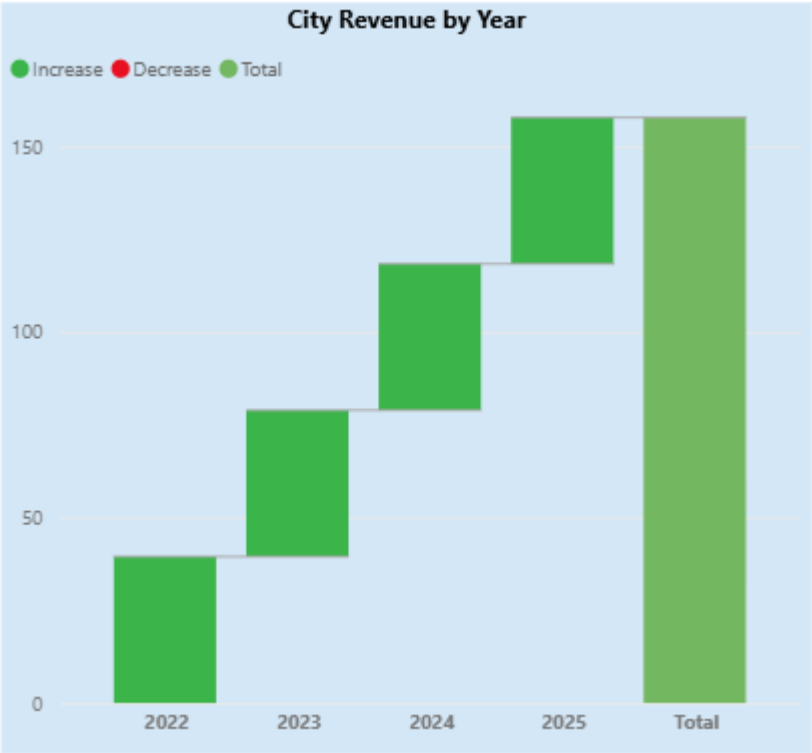


City Revenue by Year:

Using Waterfall chart,analyse the city revenue over year.



Available bikes by city:

To find out the highest and lowest available bikes by using clustered column chart.

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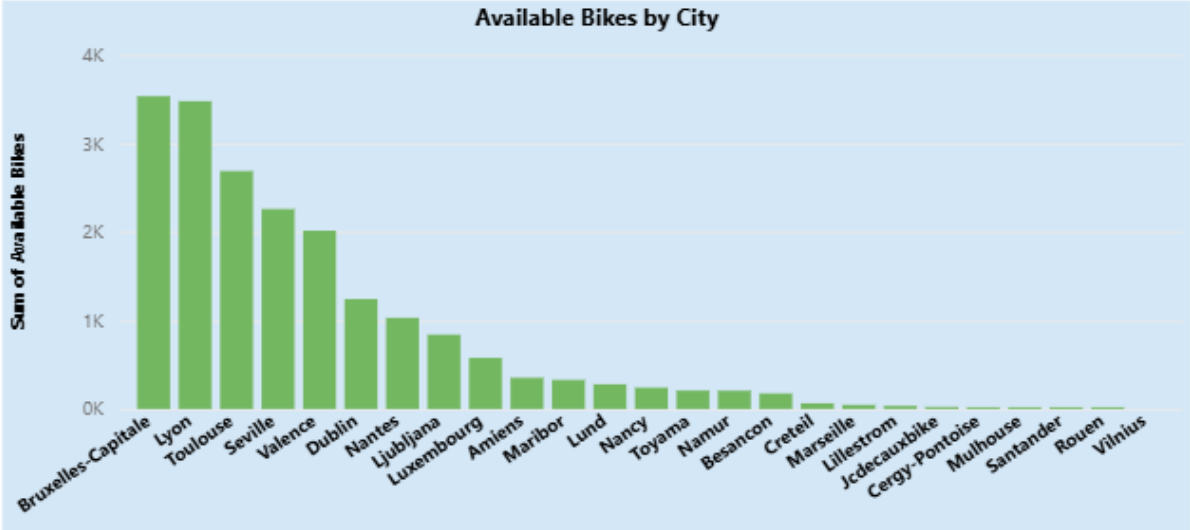
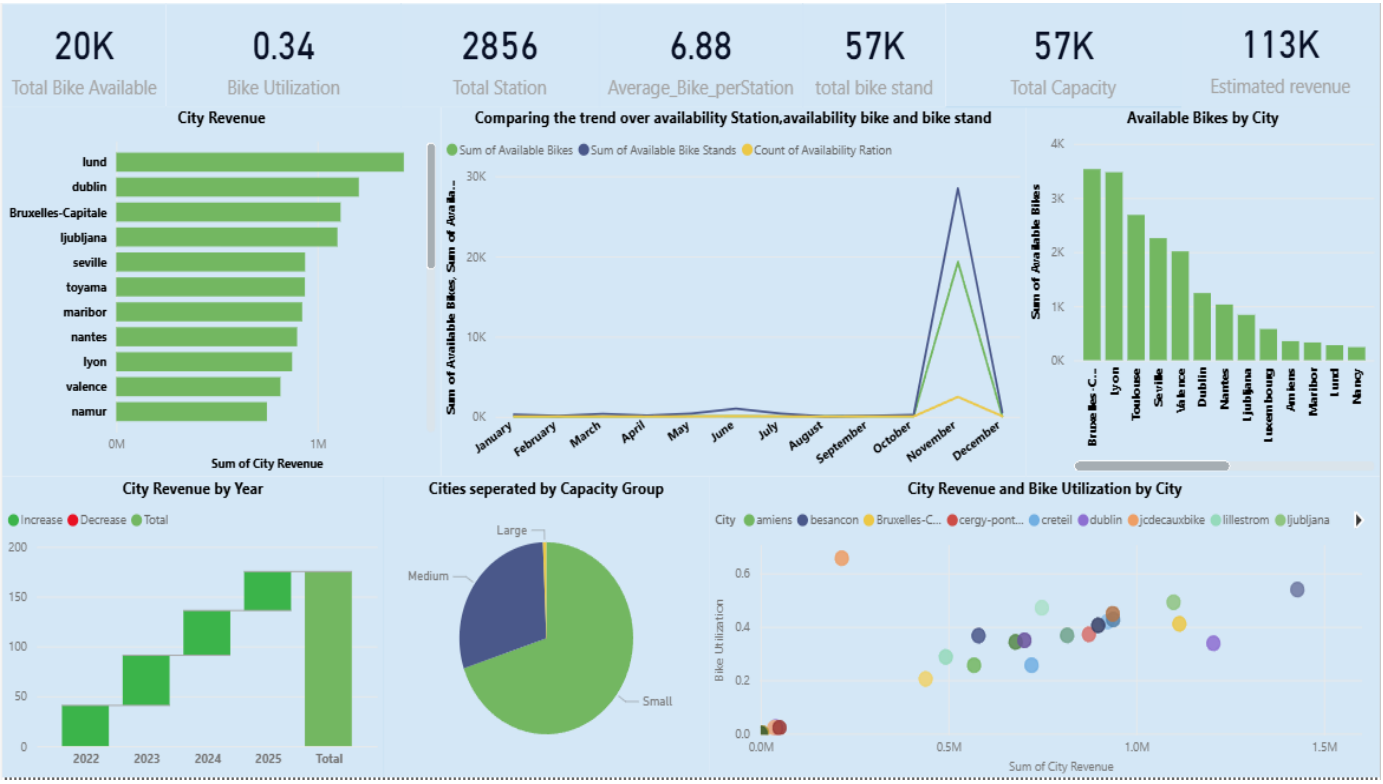


Report:



6.Insights:

Key Findings:

Analysis of **revenue efficiency by city** shows that some cities generate significantly more revenue per kilometer than others.

City level performance differences indicate that **urban mobility patterns vary across cities**, influenced by population density, commuting behavior, and infrastructure.

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Several stations have **high capacity but relatively low bike usage**, which means resources may not be fully utilized.

Most stations are operational (**open status**), but a small percentage may appear **closed or inactive**, which can affect overall service availability and user accessibility.

Analysis Findings:

The bike-sharing network includes 2,856 stations with a total capacity of approximately 57K bike stands. Currently, about 20K bikes are available, indicating an overall bike utilization rate of 0.34 (34%).

Cities such as **Lund, Dublin, and Bruxelles-Capitale** generate the **highest revenue levels**, indicating strong demand and frequent bike usage.

Cities including **Valence and Namur** show comparatively **lower revenue performance**, suggesting reduced bike usage or lower demand.

The monthly trend chart shows **relatively stable bike availability during most months**, with a **significant spike in November**.This spike suggests either **seasonal demand changes or operational redistribution of bikes**.

Stations fall under the **small capacity group**, while fewer stations belong to **medium or large categories**.

The yearly revenue analysis shows a **steady and consistent increase from 2022 to 2025**, indicating growing adoption of the bike-sharing service.

Higher revenue and bike availability in cities like **Dublin and Brussels-Capital** likely result from

- Higher population density
- Greater tourist activity
- Well-developed cycling infrastructure

These factors contribute to **higher bike demand and revenue generation**.

The **sharp increase in bike availability in November** may indicate:

- Seasonal redistribution of bikes
- Operational adjustments before winter
- Temporary demand fluctuations

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The **City Revenue by Year** chart shows a **steady increase from 2022 to 2025**, indicating that the bike-sharing system is likely to **continue generating higher revenues in the coming years**.

Cities currently showing moderate usage may see **increased bike adoption** if infrastructure improvements or promotional initiatives are implemented.

Stations with consistently low utilization should be recommended for relocation,capacity reduction and promotional initiatives to increase usage.

Cities generating higher revenue should consider **expanding station capacity** or installing additional bike stations to accommodate growing demand.

7.Conclusion:

The bike-sharing system demonstrates **steady growth in revenue and strong demand in several major cities**. However, the analysis also reveals **uneven utilization across stations and cities**, suggesting opportunities to optimize resource allocation. By implementing **better bike redistribution strategies, expanding high-demand stations, and improving infrastructure**, the system can achieve higher efficiency and maximize both user satisfaction and revenue generation.

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